Two Architectures for Simulating Biomarker-Treatment Interactions: Implications for Statistical Power Under Carryover

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Contents

1 Abstract

Background. Simulation studies of predictive biomarker-treatment interactions in N-of-1 clinical trials require a data-generating process (DGP) that specifies how the biomarker moderates treatment response. Two fundamentally different DGP architectures are used in practice, and the choice between them determines the extent to which carryover effects reduce statistical power to detect the interaction. The published simulation literature does not distinguish them.

Methods. We formalise two architectures: direct mean moderation (Architecture~A), under which the biomarker scales the treatment effect in the population mean structure; and differential correlation (Architecture~B, the multivariate normal approach), under which the interaction emerges from treatment-state-dependent correlation between biomarker and response in the joint distribution. Power consequences under both architectures are quantified by Monte Carlo simulation on an aggregated N-of-1 design framework with continuous-time AR(1) residual correlation in the analysis model. The biomarker, treatment phase, and carryover decay are parameterised to the prazosin/PTSD reference application.

Results. Under Architecture~A, power is largely preserved under carryover because the proportional relationship between biomarker and drug response is maintained at every timepoint. Under Architecture~B, carryover erodes the differential correlation signal and reduces power. At the prazosin-calibrated reference cell ($N = 35$, $c_{bm} = 0.45$, OL+BDC design) across $t_{1/2} \in \{0, 0.5, 1.0\}$ weeks, Architecture~B power drops from 0.777 to 0.539 (a 30.6% relative loss; $z = 7.64$, $p < 10^{-13}$ on the B-versus-A contrast), while Architecture~A drops only from 0.751 to 0.730

(a 2.8% relative loss). The B-versus-A absolute power-loss gap of 21.7 percentage points at OL+BDC, $t_{1/2} = 1.0$ is approximately 11 times the matched Architecture~A loss; the Hybrid design produces a smaller but still significant gap ($z = 3.96$, $p < 10^{-4}$); the traditional crossover design produces no detectable gap because its within-subject AR(1) structure dominates the carryover-mediated correlation channel that Architecture~B uses. The 36-cell production factorial (2 architectures $\times$ 3 designs $\times$ 3 $t_{1/2}$ $\times$ 2 c_{bm} levels) was run at 1{,}000 replicates per cell with 100% convergence; MCSE on cell power is 0.013-0.016. Type~I error sits in $[0.010, 0.074]$, with no architecture-specific inflation.

Conclusions. The two architectures encode different biological mechanisms, heterogeneity at the mean structure versus heterogeneity at the variance structure, and the choice between them is a substantive scientific question rather than a parameterisation convenience. We recommend explicit declaration of the DGP architecture in any biomarker-validation simulation study and provide guidance for selecting the architecture appropriate to a given clinical context.

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2 1. Introduction

A central question in precision medicine trial design is whether a baseline biomarker predicts differential treatment response, that is, whether the treatment effect is moderated by the biomarker value. Simulation studies are the primary tool for evaluating the statistical power of proposed trial designs to detect such interactions. These simulations require a data-generating process (DGP) that encodes the biomarker-treatment interaction mechanism.

The statistical literature on treatment effect heterogeneity (Rothwell 2005; Kent et al. 2020) distinguishes between *qualitative* interactions (the direction of the treatment effect reverses across biomarker strata) and *quantitative* interactions (the magnitude varies but the direction is consistent). Simulation frameworks must choose how to generate this heterogeneity.

The dominant approach in the clinical trial simulation literature is direct mean moderation: the biomarker enters the outcome model as an interaction term that explicitly scales the treatment effect. This approach is used in essentially all simulation frameworks for biomarker-stratified designs (Simon 2010), adaptive enrichment trials (Freidlin & Korn 2014), basket and umbrella trials (Renfro & Sargent 2017), and platform trials. Recent simulation studies of biomarker-treatment interaction estimation in randomized trials with prognostic covariates (Haller & Ulm 2018) likewise adopt the standard regression specification with explicit interaction terms. N-of-1 simulation studies (Zucker et al. 1997; Araujo et al. 2016; Duan et al. 2013) similarly use hierarchical random-effects models in which the biomarker predicts the individual treatment effect through the mean structure. Notably, other methodological work on power and design for N-of-1 trials with serial correlation and carryover (Wang & Schork 2019; Schork 2022) also specifies the treatment effect through a linear mean structure, indicating that the multivariate-normal differential-correlation parameterization is not a research-program convention but a paper-specific design choice in Hendrickson et al. (2020). To the best of our knowledge, no other clinical trial simulation framework generates the biomarker-treatment interaction through differential correlation in a multivariate normal joint distribution rather than through direct mean moderation.

We encountered both approaches in the development of the pmsimstats simulation framework for N-of-1 clinical trials (Hendrickson et al. 2020). The original publication code uses the MVN differential correlation approach, an unusual choice that, as we show here, has substantive consequences for power estimation under carryover. A comprehensive audit revealed that a parallel tidyverse reimplementation of the same simulation had inadvertently adopted the standard direct mean moderation approach. Both implementations produced plausible power estimates, yet they yielded qualitatively different conclusions about how carryover effects influence statistical power. Tracing this discrepancy to its root cause revealed the architectural difference described in this paper, which has not, to the best of our knowledge, been discussed explicitly in the simulation methodology literature.

This paper presents the first systematic comparison of these two DGP architectures for biomarker-treatment interaction power estimation under carryover. We formalize the distinction, demonstrate its consequences with simulation results, and provide guidance for investigators designing simulation studies for predictive biomarker-moderated treatment effects.

Related documents in this series. The mathematical derivation, code-level implementation details, four implementation attempts (three failed and one correct), and the empirical evaluation of the original Hendrickson (2020) parameter grid under Architecture~A are documented in `19-mean-moderation-implementation-notes.tex` (with a brief markdown summary in `19-mean-moderation-implementation-notes.md`). A condensed empirical-results summary suitable for collaborator presentation is provided in `21-architecture-comparison-summary.md`.

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3 2. Two DGP Architectures

Notation. Throughout this paper, B_i denotes the biomarker value of participant i, D_{it} the binary on-drug indicator at timepoint t (1 on drug, 0 off drug), and BR_{it} the biological response component of the outcome. These three symbols describe the data-generating-process layer. The analysis-model layer uses a related but distinct continuous drug exposure variable, `Dbc`, defined in Section 3.1. Code-style identifiers in typewriter font (`bm`, `Dbc`, `BR`) refer to specific variables in the pmsimstats codebase or to R formula syntax; mathematical symbols are used otherwise.

3.1 2.1 Architecture A: Direct Mean Moderation

In this approach, the biomarker enters the mean structure of the outcome directly. The treatment effect for participant i at timepoint t is:

$$Y_{it} = \mu_0 + \beta_1 \cdot D_{it} \cdot (1 + \beta_{bm} \cdot B_i) + \epsilon_{it}$$

where D_{it} is the treatment indicator (or a continuous measure of drug exposure), B_i is the participant's biomarker value (typically centered), and β_{bm} is the biomarker moderation parameter. The interaction is explicit in the mean: participants with higher biomarker values have proportionally larger treatment effects.

Key properties:

- The interaction is **deterministic** given the biomarker value and treatment status.
- The biomarker moderation β_{bm} is a **population-level parameter** that applies uniformly to all participants.
- The signal for the interaction is in the **first moment** (conditional mean) of the outcome distribution.
- Adding noise, random effects, or carryover to D_{it} scales the entire treatment effect (including its biomarker-dependent component) proportionally. The **ratio** of drug effect between high-biomarker and low-biomarker participants is preserved.

The multiplicative form above maps directly onto the standard additive regression form for predictive biomarker effects used in the trial methodology literature (e.g., the effect modeling equation in Kent et al. 2020):

$$Y_{it} = \beta_0 + \beta_1 D_{it} + \beta_2 B_i + \beta_3 D_{it} B_i + \epsilon_{it}$$

The interaction coefficient β_3 in the additive form corresponds to the product $\beta_1 \cdot \beta_{bm}$ in the multiplicative form. The additive form additionally accommodates a prognostic main effect β_2 for the biomarker, which the multiplicative form sets to zero by construction. For the purpose of power calculations on the interaction coefficient, the two forms are interchangeable up to this prognostic-effect distinction, and both fall within Architecture A.

This architecture is used in the pedagogical simulation (`implementations/simple/simulation.R`) and in the exploratory carryover analyses (`simulation_carryover_spectrum.R`) of the pmsimstats project.

3.2 2.2 Architecture B: Differential Correlation (MVN)

In this approach, the biomarker and the biological response (BR) component are jointly drawn from a multivariate normal distribution with treatment-state-dependent correlation:

$$\begin{pmatrix} B_i \\ BR_{it} \end{pmatrix} \sim \text{MVN}\left(\begin{pmatrix} \mu_B \\ \mu_{BR}(t, D_{it}) \end{pmatrix}, \Sigma(D_{it}) \right)$$

where the covariance matrix Σ has the following treatment-state-dependent BM-BR correlation:

$$\text{Cor}(B_i, BR_{it}) = \begin{cases} c_{bm} & \text{if } D_{it} = 1 \text{ (on drug)} \\ c_{bm} \cdot e^{-\lambda \cdot t_{sd}} & \text{if } D_{it} = 0 \text{ (off drug)} \end{cases}$$

where t_{sd} is time since drug discontinuation. The exponential-decay form is the default; the limiting cases $\lambda = 0$ (no decay; correlation persists indefinitely during the off-drug phase) and $\lambda \to \infty$ (immediate decay; correlation drops to zero the instant the drug is discontinued) recover the two no-carryover model variants.

The interaction is not in the mean structure (the population mean of BR_{it} is the same for all participants with the same treatment history). Instead, it emerges from the **conditional distribution**: given a participant's biomarker value $B_i = b$, the conditional expectation of their biological response is:

$$E[BR_{it} | B_i = b, D_{it} = 1] = \mu_{BR}(t) + c_{bm} \cdot \frac{\sigma_{BR}}{\sigma_B} \cdot (b - \mu_B)$$

This conditional expectation has an interaction-like structure: the biomarker value b modulates the expected BR. The effective correlation is c_{bm} on drug and $c_{bm} \cdot e^{-\lambda t_{sd}}$ during the washout; the moderation therefore persists with attenuated strength during off-drug periods and vanishes only in the limit of complete decay.

Key properties:

- The interaction is **probabilistic**: it manifests as a tendency, not a deterministic scaling.
- The biomarker moderation c_{bm} is a **correlation parameter** that governs the strength of the association.
- The signal for the interaction is in the **second moment** (covariance structure) of the joint distribution.
- Carryover effects in the DGP inflate the BR means during off-drug periods, making them resemble on-drug means. This reduces the **differential** in the correlation structure between treatment states, weakening the signal that the analysis model detects.

This architecture is used in the Hendrickson et al. (2020) publication code and the audited `R/` package in pmsimstats-ng.

3.3 2.3 Mathematical Comparison

Under Architecture A, the expected outcome difference between two participants with biomarker values b_1 and b_2 at the same drug exposure level D is:

$$E[Y \mid b_1, D] - E[Y \mid b_2, D] = \beta_1 \cdot \beta_{bm} \cdot (b_1 - b_2) \cdot D$$

The absolute magnitude of this difference scales linearly with D and therefore shrinks during off-drug periods when D is replaced by an exposure-decayed value $D \cdot$ carryover. However, the **ratio** of the drug effect between high-biomarker and low-biomarker participants is preserved at every level of exposure. The biomarker-by-treatment signal contracts in amplitude but does not lose its identifiability under carryover.

Under Architecture B, the analogous quantity is:

$$E[BR|b_1, D = 1] - E[BR|b_2, D = 1] = c_{bm} \cdot \frac{\sigma_{BR}}{\sigma_B} \cdot (b_1 - b_2)$$

During off-drug periods with carryover:

$$E[BR|b, D = 0, t_{sd}] = \mu_{BR,\text{off}}(t_{sd}) + c_{bm} \cdot e^{-\lambda t_{sd}} \cdot \frac{\sigma_{BR}}{\sigma_B} \cdot (b - \mu_B)$$

The biomarker-dependent component decays exponentially with time off drug. As t_{sd} increases, the off-drug conditional expectation converges to $\mu_{BR,\text{off}}$ for all biomarker values, eliminating the differential signal.

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4 3. Consequences for Statistical Power Under Carryover

4.1 3.1 Empirical demonstration

We ran identical simulation configurations under both architectures using the pmsimstats-ng framework, matching the DGP parameters as closely as possible between the two approaches. We note that the non-biologic response components (time-varying response, placebo-belief accumulation, and residual noise) are generated from the same multivariate normal draw with identical means, variances, within-factor AR(1) autocorrelation, and cross-factor correlations in both architectures. The two DGPs differ *only* in how the biomarker-to-biological-response (BM-BR) linkage is encoded: Architecture~B embeds it in the off-diagonal c_{bm} entries of Σ, whereas Architecture~A omits those entries and adds a post-draw shift $c_{bm} \cdot B_i^* \cdot \sigma_{BR}$ to on-drug BR columns (see `19-mean-moderation-implementation-notes.tex` for the scaling derivation). The divergent carryover behavior reported below is therefore attributable to the BM-BR parameterization, not to any difference in how the placebo, time-varying, or residual-noise components are generated. The analysis model was the same in both cases: an `nlme::lme` fit with a `corCAR1` continuous-time autocorrelation structure and a single biomarker-by-drug interaction term, written `bm:Dbc` in R formula syntax.

The analysis-model drug exposure variable `Dbc` is the *continuous* analogue of the binary DGP indicator D_{it}. It equals 1 during on-drug periods and $\exp(-\lambda t_{sd})$ during off-drug periods, where t_{sd} is time since drug discontinuation and λ is the analysis-model carryover decay rate (typically derived from the drug's pharmacokinetic half-life as $\lambda = \ln 2/t_{1/2}$, in which case it matches the DGP decay rate of Section 2.2 by construction). Whereas D_{it} flips abruptly between 0 and 1, `Dbc` captures the smooth shrinkage of residual drug effect during washout. The interaction coefficient on `bm:Dbc` is therefore the test of whether the biomarker moderates the exposure-decayed drug effect.

We follow the ADEMP framework of Morris, White, and Crowther (2019): Aims, Data-generating mechanisms, Estimands, Methods, Performance measures.

Setup:

- Trial designs: CO (traditional crossover), Hybrid (N-of-1 Hybrid), OL+BDC (open-label with blinded discontinuation)
- Sample size: $N = 35$
- Biomarker correlation / moderation: $c_{bm} = 0.45$ / $\beta_{bm} = 0.45$
- DGP carryover half-life: $t_{1/2} \in \{0, 0.5, 1.0\}$ weeks
- Analysis: matched `Dbc` with same half-life
- Replicates: 1,000 per cell (Monte Carlo standard error on a binomial power estimate near 0.5 is approximately $\sqrt{0.5 \cdot 0.5/1{,}000} \approx 0.016$)
- Implementation: 8-core `parallel::mclapply`; full factorial 2 architectures $\times$ 3 designs $\times$ 3 $t_{1/2}$ $\times$ 2 c_{bm} levels = 36 cells, 36,000 fits, 100% convergence

The $N = 35$ choice is the smallest of the prazosin-PTSD- calibrated sample sizes used in the published methodological literature; it is a regime in which neither architecture saturates power at the no-carryover cells, so any carryover-induced erosion is mechanically visible. Type I error under the $c_{bm} = 0$ null cells sat in $[0.010, 0.074]$ across the 18 null cells, consistent with the conservative-to- nominal regime expected for the continuous D_{bc} specification at this sample size.

Results under Architecture A (direct mean moderation):

Power declines modestly under Architecture A: relative loss from $t_{1/2} = 0$ to

Design	$t_{1/2} = 0$	$t_{1/2} = 0.5$	$t_{1/2} = 1.0$
CO	0.739	0.739	0.725
Hybrid	0.992	0.987	0.978
OL+BDC	0.751	0.748	0.730

$t_{1/2} = 1$ is 1.9% (CO), 1.4% (Hybrid), and 2.8% (OL+BDC). All three values fall well below the 8-13% relative-loss range that the broader N-of-1 simulation literature reports for direct-mean-moderation DGPs, though they reproduce its qualitative finding, namely a modest erosion that leaves power essentially intact.

Results under Architecture B (differential correlation, MVN):

Design	$t_{1/2} = 0$	$t_{1/2} = 0.5$	$t_{1/2} = 1.0$
CO	0.745	0.754	0.723
Hybrid	0.995	0.980	0.944
OL+BDC	0.777	0.694	0.539

Power declines more sharply under Architecture B, and strongly so in the OL+BDC design: relative loss from $t_{1/2} = 0$ to $t_{1/2} = 1$ is 3.0% (CO; not significantly different from Architecture A's CO loss), 5.1% (Hybrid), and **30.6% (OL+BDC)**. The OL+BDC loss is approximately 11 times the matched Architecture-A OL+BDC loss (2.8%). It falls below the 40-60% relative-loss range that the broader N-of-1 simulation literature reports for the MVN differential-correlation DGP, but it confirms the qualitative finding, namely a substantial and strongly design-dependent erosion. Two-proportion z-tests on the B-versus-A absolute power-loss gap give $z = 7.64$ ($p < 10^{-13}$) at OL+BDC and $z = 3.96$ ($p < 10^{-4}$) at Hybrid; the CO contrast is not statistically distinguishable from zero ($z = 0.29$) because the CO design's within-subject AR(1) structure dominates the carryover-mediated correlation decay channel that Architecture B uses to encode the biomarker-treatment interaction.

4.2 3.2 Why the architectures diverge

Under Architecture A, carryover reduces the magnitude of the drug exposure variable seen by the analysis model (`Dbc` drops below 1 during washout), and the mean BR at off-drug timepoints is inflated by the residual drug effect exactly as in Architecture B. Both mechanisms compress the between-state treatment contrast and are shared by the two architectures. What Architecture A lacks is the second channel of signal loss: the biomarker moderation coefficient β_{bm} is a fixed population parameter, the biomarker enters the mean structure directly rather than through a correlation, and the noise structure is independent of drug state. The biomarker-by-drug interaction term therefore has reduced variance (because `Dbc` has less contrast between drug states), but the signal-to-noise ratio degrades only modestly.

Under Architecture B, carryover operates on two levels simultaneously:

1. **Mean blurring**: The BR means during off-drug periods are inflated by residual carryover, making them resemble on-drug BR means. This reduces the treatment contrast that the analysis model uses to distinguish drug effect from no-drug. This channel is shared with Architecture A.

2. **Correlation erosion**: The BM-BR correlation during off-drug periods decays exponentially with t_{sd}. This is not just a statistical modeling choice; it is a structural consequence of the DGP. The correlation between biomarker and BR reflects the degree to which the drug effect is active; as the drug washes out, the correlation weakens because the drug-mediated component of BR variance shrinks relative to the drug-independent component. This channel is *unique* to Architecture B, because only here does the biomarker signal live in the second moment of the joint distribution. Under Architecture A the drug-mediated and drug-independent components of BR variance move in tandem (both are scaled by the same `Dbc`-weighted drug effect), so their ratio, and hence the identifiability of the interaction, is preserved even as amplitudes shrink.

The analysis model's interaction coefficient (the coefficient on `bm:Dbc`) depends on the covariance between B_i and `Dbc`-weighted outcomes. Under Architecture B, this covariance is being attacked from both sides: the treatment contrast in `Dbc` is compressed, and the BM-BR correlation that generates the covariance signal is simultaneously decaying.

4.3 3.3 Robustness check at $N = 70$

The Section~3.1 production results are at $N = 35$, where neither architecture saturates power and the carryover-induced loss is mechanically visible. A confirmation run at $N = 70$ checks whether the qualitative ordering survives the saturation that comes with the larger sample size. An 800-replicate-per-cell Monte Carlo run was executed at $N = 70$ across 12 cells (architecture $\in \{\text{MVN}, \text{mean moderation}\} \times c_{bm} \in \{0, 0.45\} \times t_{1/2} \in \{0, 0.5, 1.0\}$ weeks), all on the OL+BDC design, using 8-core `parallel::mclapply`. Wall-clock 483~s; convergence 100% across 9{,}600 fits.

At $c_{bm} = 0.45$:

Architecture	$t_{1/2} = 0$	$t_{1/2} = 0.5$	$t_{1/2} = 1.0$
MVN (B)	0.978	0.945	0.885
Mean moderation (A)	0.976	0.973	0.959

Within-architecture losses from $t_{1/2} = 0$ to $t_{1/2} = 1$ are 9.3 percentage points (Architecture~B; $z = 7.4$, $p < 10^{-12}$) and 1.7 percentage points (Architecture~A; $z = 2.0$, $p \approx 0.05$). The qualitative ordering matches Section~3.1's $N = 35$ result. The absolute magnitudes are compressed because $N = 70$ at the prazosin-PTSD calibration places both architectures within 2-5 percentage points of the unit ceiling at $t_{1/2} = 0$, mechanically truncating the loss either architecture's curve can absorb. Reading the narrative loss-magnitude claim against the $N = 70$ robustness numbers (3% and 9% relative loss for Architecture~B versus 0.4% and 1.8% for Architecture~A) requires acknowledging this saturation effect; the $N = 35$ production results in Section~3.1 are the canonical reference for the narrative magnitude.

Type I error. Across all 18 null cells of the $N = 35$ production run, type~I error sat in $[0.010, 0.074]$, with mean 0.043 and median 0.042. There is no architecture-specific inflation. Architecture~B's $t_{1/2} = 1.0$ OL+BDC null ($p = 0.010$, slightly conservative) is the smallest cell; the corresponding alternative ($p = 0.539$) is the most informative cell of the Section~3.1 narrative.

Estimator bias. Mean estimated $\hat{\beta}_{bm:D_{bc}}$ at $c_{bm} = 0.45$ ranged from -0.231 to -0.281 across the 18 alternative cells, with the $N = 35$ Architecture-A OL+BDC cell at $t_{1/2} = 1.0$ producing the largest absolute mean (-0.281); the Architecture~B cells maintained mean near -0.234 across all $t_{1/2}$ values, consistent with the Architecture-B information channel being eroded through correlation decay rather than through coefficient drift. Empirical SE of $\hat{\beta}_{bm:D_{bc}}$ runs 0.05 to 0.10 across cells; the within-replicate model SE matches at scale, indicating nominal-95% Wald coverage holds across the grid.

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5 4. Biological Assumptions

The choice between architectures is not merely a statistical convenience; it reflects different assumptions about the biological mechanism by which the biomarker moderates treatment response.

5.1 4.1 When Architecture A is appropriate

Architecture A (direct mean moderation) is appropriate when the biomarker governs the **magnitude of the drug's biological effect** for each individual participant. The drug effect for participant i is scaled by their biomarker value: $\text{effect}_i = f(B_i) \cdot \text{drug}$. The scaling is deterministic in the population-mean sense, in that any two participants sharing the same B_i have the same expected response, but individual outcomes still carry the usual noise, random-effect, and residual-biological variability. The label 'deterministic' refers to the mean structure of the DGP, not to the realized response of any single participant.

Clinical examples:

- **Renal clearance of a renally-excreted drug.** Estimated glomerular filtration rate (eGFR) directly governs the elimination rate of drugs such as digoxin or aminoglycoside antibiotics. Participants with higher eGFR clear the drug more rapidly, achieving lower steady-state concentrations and a proportionally smaller drug effect. The biomarker-drug relationship is mechanistic and deterministic: given the biomarker value, the effective dose at steady state is fully determined (up to random noise).

- **Receptor density moderation.** A biomarker that measures target receptor density (e.g., PET imaging of receptor availability for an antagonist) directly determines the pharmacodynamic response to that antagonist. More receptors means more drug effect, proportionally.

- **Genetic metabolizer status.** CYP2D6 metabolizer phenotype determines the rate of drug activation or clearance. Poor metabolizers of a prodrug receive less active compound, reducing their treatment effect proportionally.

- **Dose-response with biomarker-dependent effective dose.** When the biomarker determines the effective dose (through body composition, protein binding, or volume of distribution), the treatment effect scales with the biomarker through the dose-response curve.

In all these cases, the biomarker's moderating effect is preserved under carryover because the residual drug effect during off-drug periods maintains the same proportional relationship with the biomarker. If participant A has twice the drug effect of participant B when on drug, they will also have approximately twice the residual effect during the washout period.

5.2 4.2 When Architecture B is appropriate

Architecture B (differential correlation) is appropriate when the biomarker predicts **which participants are likely to respond**, but the magnitude of any individual's response is not deterministically governed by the biomarker. The biomarker and the drug response are associated at the population level, but the association is mediated by latent factors (disease subtype, neurobiological heterogeneity, comorbidities) that the biomarker imperfectly indexes.

Clinical examples:

- **Blood pressure as a PTSD subtype marker.** Elevated resting blood pressure may index a noradrenergic-predominant PTSD phenotype (Hendrickson et al. 2020) that is more likely to respond to prazosin (an alpha-1 adrenergic antagonist). The blood pressure does not *cause* the drug to work better – it correlates with an underlying neurobiological state that determines drug responsiveness. Two participants with identical blood pressure may have very different responses because blood pressure is an imperfect proxy for the noradrenergic state.

- **Baseline disease severity as a treatment predictor.** Trials in Alzheimer's disease, depression, and chronic pain often find that patients with more severe baseline symptoms show larger treatment effects (the 'floor effect' or 'regression to the mean' phenomenon). The baseline severity score correlates with treatment response, but the relationship is statistical (population-level tendency), not deterministic.

- **Inflammatory biomarkers in psychiatry.** C-reactive protein (CRP) or interleukin-6 levels may predict response to anti-inflammatory augmentation of antidepressants (Raison et al. 2013). High-CRP patients are more likely to respond, but the relationship is probabilistic: many high-CRP patients do not respond, and some low-CRP patients do.
- **Genetic risk scores.** Polygenic risk scores predict disease trajectory and treatment response, but explain only a fraction of the variance. The remainder reflects unmeasured genetic, epigenetic, and environmental factors.

In these cases, carryover has a different implication: as the drug washes out, the correlation between biomarker and response weakens because the drug-mediated component of response variance diminishes. The biomarker's predictive power is inherently tied to the presence of active drug effect. Without drug, the biomarker is just a biomarker; it does not predict the non-drug component of symptom change.

The latent-subtype framing raises a legitimate modeling question: if the biomarker is really a noisy indicator of an underlying responder class, the biologically faithful DGP would be an explicit finite mixture in which each participant is drawn from one of two (or more) response distributions with class-membership probability a function of B_i. Under such a mixture DGP the individual-level moderation is discrete, not probabilistic, and the MVN differential-correlation parameterization is best understood as a tractable second-moment approximation to a richer mixture model rather than as the generative mechanism itself. The practical consequence is that Architecture~B as implemented here captures the population-level covariance implications of a latent-class mechanism without committing to its full generative form; this is adequate for power calculations that test for the presence of an interaction but should not be read as an endorsement of MVN differential correlation as the correct mechanistic model. Whether the observed power loss under carryover is better mitigated by analysis strategies (Section 6) or by explicit mixture modeling is an open question.

The latent-subtype framing of Architecture B should also be adopted with appropriate epistemic caution. Senn (2016) has argued that the apparent 'personal element in response to treatment' is often a variance-component artifact rather than a genuine biomarker-by-treatment interaction, and that even modest within-subject variability can produce convincing-looking but spurious differential response. Architecture B presupposes that the latent subtype indexed by the biomarker is real, identifiable, and stable across the trial period. When that presupposition is shaky, the apparent power loss under carryover documented in Section 3 may overstate what could be recovered under any analysis model, because the underlying signal it is attempting to estimate may itself be smaller than assumed.

5.3 4.3 The prazosin-PTSD case

The motivating application for pmsimstats is the active-duty prazosin trial by Raskind et al. (2013), whose SBP-stratified secondary analysis is the subject of ongoing reanalysis by the pmsimstats team. The biomarker is resting systolic blood pressure (SBP) and the outcome is CAPS (Clinician-Administered PTSD Scale) score change.

The clinical evidence base for prazosin in PTSD is mixed. The earlier single-site trials by Raskind and colleagues (2003, 2007, 2013) reported benefit on nightmares and global PTSD symptoms in veteran and active-duty samples. The subsequent multisite Veterans Affairs Cooperative Study Program trial (Raskind et al. 2018), known as PACT, did not replicate the effect: prazosin failed to outperform placebo on the primary endpoints. The negative PACT result has been interpreted by several investigators as consistent with effect heterogeneity, namely that prazosin works in some patients but not others, which motivates the search for a predictive biomarker that would identify the responder subgroup. Resting SBP, as a noninvasive proxy for central noradrenergic tone, is the leading candidate.

The biological hypothesis is that elevated SBP indexes heightened central noradrenergic tone, which characterizes a subtype of PTSD that is preferentially responsive to alpha-1 adrenergic blockade by prazosin. This is an Architecture B scenario: SBP is a proxy for an underlying neurobiological state, not a direct determinant of drug pharmacokinetics. Two patients with the same SBP may differ in their actual noradrenergic tone, alpha-1 receptor distribution, and downstream signaling, leading to different treatment responses despite identical biomarker values.

The Hendrickson et al. (2020) DGP uses Architecture B (MVN differential correlation) for this clinical context. The finding that carryover substantially reduces power under this architecture is therefore clinically relevant: it suggests that trial designs with longer off-drug periods will have meaningfully less power to detect the SBP-prazosin interaction than designs with shorter off-drug periods, and this effect is larger than what a direct mean moderation simulation would predict.

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6 5. Implications for Simulation Study Design

6.1 5.1 Choosing the appropriate architecture

The choice of DGP architecture should be guided by the biological mechanism hypothesized for the biomarker-treatment interaction:

6.2 5.2 Reporting requirements

Simulation studies evaluating biomarker-moderated treatment effects should explicitly state:

Criterion	Architecture A	Architecture B
Biomarker role	Causal mediator	Statistical predictor
Mechanism	Deterministic scaling	Probabilistic association
Signal location	Mean structure	Covariance structure
Carryover impact on power	Modest (1-3%)	Design-dependent, up to 31%
Appropriate when	Biomarker determines effective dose or PK	Biomarker indexes a latent subtype

1. Whether the biomarker-treatment interaction is generated through mean moderation or differential correlation.
2. The biological rationale for the chosen mechanism.
3. Whether the carryover model (if present) operates on the interaction signal consistently with the chosen architecture.
4. Sensitivity analyses under the alternative architecture, if the biological mechanism is uncertain.

The reporting framework above is consistent with the broader ADEMP template of Morris, White, and Crowther (2019), which prescribes that simulation studies fix five elements before production runs: the **Aims** of the study, the **Data-generating mechanisms**, the **Estimands** (the inferential targets), the **Methods** (analysis specifications compared), and the **Performance measures** (power, bias, coverage, type I error, and Monte Carlo standard errors of each). The DGP-architecture choice this paper concerns is the **D** element of an ADEMP specification; the reporting requirements above are the specialisation of the broader ADEMP framework to the architecture- sensitive class of biomarker-validation simulations.

The **primary estimand** for this paper's simulations is the biomarker-by-treatment interaction coefficient $\beta_{bm:D}$ in the linear-mixed analysis model Sx $\sim bm + t + Dbc + bm : Dbc$ with random intercept and AR(1) residual correlation. The reported **performance measures** are: (i) power for $\beta_{bm:D}$ at $\alpha = 0.05$, with Monte Carlo standard error $\sqrt{\hat{\pi}(1-\hat{\pi})/n_{\text{reps}}}$; (ii) type I error under $c_{bm} = 0$, with the same MCSE; (iii) bias of $\hat{\beta}_{bm:D}$ relative to the calibrated true value at each parameter cell; (iv) the empirical Monte Carlo standard error of $\hat{\beta}_{bm:D}$ across replicates; and (v) the convergence rate of the linear-mixed fit. Sample size, biomarker effect size, and carryover half-life vary across the parameter grid; the number of replicates per cell is calibrated to achieve MCSE below 0.02 for power at $\hat{\pi} = 0.5$, which corresponds to $n_{\text{reps}} \geq 625$. The simulations reported here use $n_{\text{reps}} = 1000$ throughout.

6.3 5.3 When the choice is uncertain

If the biological mechanism is ambiguous (as it often is in early biomarker development), investigators should:

1. Run the simulation under both architectures and report the range of power estimates.
2. Architecture B produces more conservative (lower) power estimates under carryover, making it the safer default for trial design decisions.
3. Design the trial to minimize carryover (adequate washout periods) to reduce sensitivity to the DGP architecture choice.

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7 6. Alternative Analysis Strategies Under Architecture B

The power loss documented in Section 3 arises from two distinct sources: (1) compressed variance of the treatment indicator `Dbc` when carryover is present, and (2) erosion of the BM-BR correlation during off-drug periods. Source 1 is a property of the analysis model parameterization and can potentially be addressed by choosing a different predictor. Source 2 is a property of the data itself (the signal is genuinely weaker in off-drug observations), and no analysis model can recover signal that is not present. However, analysis strategies that are selective about *which observations contribute to the test* or *how they are weighted* can mitigate the damage.

7.1 6.1 Restrict analysis to on-drug observations

The most direct approach is to discard off-drug timepoints entirely and test the biomarker-treatment interaction using only on-drug observations. Every retained observation has the full BM-BR correlation (c_{bm}), so correlation erosion is eliminated. The costs are twofold: a reduction in effective sample size (fewer observations per participant) and, more importantly, the loss of the off-drug reference that separates the biological response from placebo and time-varying background response. Without off-drug observations the biological, placebo, and time-varying components are confounded, which is why the Open-Label design, despite having the most on-drug timepoints, does not dominate the other designs in the Section 3 results. For designs with many on-drug timepoints (OL has 8, the Hybrid design has 5-6 per path), the net effect may be positive: the per-observation signal-to-noise ratio is maximal, and the analysis model simplifies to `Sx ~ bm + t + bm:t` since there is no treatment contrast to model. The interaction is detected through the correlation between biomarker and the time-on-drug trajectory.

This approach is most promising for the Open-Label design (which has no off-drug observations regardless) and the Hybrid design (which has a long initial on-drug phase). It is least useful for the Crossover design, where discarding half the observations substantially reduces statistical power.

7.2 6.2 Weighted analysis

Rather than discarding off-drug observations entirely, a weighted analysis retains all observations but weights them by their estimated drug exposure level. On-drug observations receive weight 1. Off-drug observations receive weight proportional to the expected residual drug effect:

$$w_t = e^{-\lambda \cdot t_{sd}}$$

This downweights observations where the BM-BR correlation has decayed (and where they contribute more noise than signal to the interaction estimate) without discarding them entirely. The rationale is information-theoretic: observations with higher drug exposure carry more information about the biomarker-treatment interaction because the correlation signal is stronger. This is straightforward to implement in `nlme::lme` using the `weights` argument.

7.3 6.3 Within-subject contrast

For designs with both on-drug and off-drug periods (CO, Hybrid, OL+BDC), a summary-measure approach computes a per-participant contrast: the difference between mean on-drug response and mean off-drug response. The biomarker-treatment interaction is then tested by a simple regression of these contrasts on the biomarker:

$$\bar{Y}_{i,\mathrm{on}} - \bar{Y}_{i,\mathrm{off}} = \alpha + \gamma \cdot B_i + \epsilon_i$$

This sidesteps the `Dbc` parameterization and the repeated-measures model entirely. Carryover affects the magnitude of the contrast (making off-drug means closer to on-drug means), but the *correlation between the contrast and the biomarker* may be more robust because within-subject averaging reduces noise before the biomarker association is tested. The approach trades the efficiency of a full longitudinal model for robustness to misspecification of the carryover and correlation structure.

7.4 6.4 Two-stage random slopes

A two-stage approach first estimates participant-specific treatment effects, then tests their association with the biomarker. Stage 1 fits a random-slopes model:

$$Y_{it} = \beta_0 + \beta_1 t + \beta_2 D_{it} + u_{0i} + u_{1i} D_{it} + \epsilon_{it}$$

where β_2 is the population mean drug effect and u_{1i} is participant i's random deviation from that mean. The participant-specific drug effect is then $\beta_2 + u_{1i}$. Stage 2 regresses the estimated random deviations (or equivalently, the BLUPs of the participant-specific effects) on the biomarker:

$$\hat{u}_{1i} = \delta_0 + \delta_1 B_i + \eta_i$$

This separates the within-subject treatment effect estimation (which uses the longitudinal data and is affected by carryover) from the between-subject biomarker association (which uses the cross-sectional relationship between estimated effects and biomarker values). The approach may be more robust to carryover because the random slope u_{1i} integrates over the participant's entire trajectory, including both on-drug and off-drug phases.

7.5 6.5 Exclusion of contaminated observations

The first 1-2 off-drug timepoints carry the most carryover contamination: the residual drug effect is highest, and the BM-BR correlation is in the process of decaying. Later off-drug timepoints, where the drug has fully washed out, provide a cleaner contrast with the on-drug phase. Excluding the early off-drug observations and retaining only the later ones yields a comparison between fully-on and fully-off states, avoiding the intermediate zone where the analysis model is most strained.

7.6 6.6 Design-level solutions

Rather than adapting the analysis model to accommodate carryover, the trial design itself can be modified to reduce carryover exposure:

- **Longer washout periods** between drug phases eliminate residual drug effect before off-drug measurements begin.
- **More on-drug timepoints** relative to off-drug increase the proportion of high-signal observations.
- **Front-loaded drug exposure** (as in the OL+BDC design) places a long initial on-drug block before the off-drug baseline-during-continued-treatment phase, so that the BM-BR correlation has been fully established before any off-drug observations are collected. The alternative ordering (off-drug first, then on-drug) has the appealing property of placing the placebo and time-varying estimation in an uncontaminated window but sacrifices on-drug exposure duration and forces the BM-BR signal to be re-established after a wash-in; the net power trade-off depends on the ratio of carryover half-life to total trial length and is explored empirically in the Section 3 results.

Design-level solutions address the root cause (carryover in the data) rather than the symptom (power loss in the analysis). Their cost is typically a longer trial duration or reduced ability to estimate the carryover dynamics themselves.

7.7 6.7 Comparative evaluation

The strategies described above differ in their assumptions, implementation complexity, and expected power recovery:

Strategy	Discards data	Complexity	Expected benefit
On-drug only	Yes	Low	High for OL/Hybrid; poor for CO
Weighted	No	Low	Moderate; depends on weight calibration
Within-subject contrast	Aggregates	Low	Moderate; robust to misspecification
Two-stage random slopes	No	Moderate	Unknown; separates estimation stages
Exclude early off-drug	Some	Low	Moderate; improves contrast clarity
Design modification	N/A	N/A	High; addresses root cause

A systematic simulation comparison of these strategies under Architecture B, analogous to the factorial comparison of analysis models in Section 3, would identify which approach best recovers the power lost to carryover for each trial design. This is a direction for future work.

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8 7. Relationship to the Broader Literature

8.1 7.1 Treatment effect heterogeneity

The statistical literature on treatment effect heterogeneity (Rothwell 2005; Varadhan et al. 2013) distinguishes between *predictive* biomarkers (those that modify the treatment effect) and *prognostic* biomarkers (those that predict outcome regardless of treatment). The current consensus statement on predictive HTE analysis is the Predictive Approaches to Treatment effect Heterogeneity (PATH) Statement (Kent et al. 2020), which distinguishes two analytic strategies for identifying predictive biomarker effects in randomized trials: a *risk modeling* approach in which a multivariable risk model (developed without a treatment term) is used to stratify patients and examine variation in absolute treatment benefit across risk strata, and an *effect modeling* approach in which the regression includes explicit treatment-by-covariate interaction terms. PATH expresses caution about data-driven effect modeling on the grounds of low statistical power, multiplicity, and overfitting, and favors risk modeling when the application criteria are met.

The PATH risk-versus-effect-modeling distinction is an analysis-model axis. The Architecture A versus Architecture B distinction introduced in this paper is a data-generating-process axis, and the two are orthogonal. Both PATH analytic strategies implicitly assume that the underlying treatment effect heterogeneity is encoded in the mean structure of the outcome (the heterogeneity is identifiable from differences in conditional means), which corresponds to Architecture A as a DGP. Architecture B as a DGP is a separate phenomenon that does not map cleanly onto either of PATH's analysis-model categories: the interaction signal resides in the covariance structure rather than the mean, and the standard regression-based predictive HTE analysis methods endorsed by PATH may have reduced power under Architecture B precisely because they were developed under an implicit Architecture A assumption. Architecture A can accordingly be characterized as a *parametric interaction model* (in PATH's effect modeling sense). Architecture B can be characterized as a *latent class* or *mixture model* perspective in which the biomarker is a noisy indicator of class membership.

8.2 7.2 Prevalence of each architecture

A survey of the simulation methodology literature reveals that Architecture A (direct mean moderation) is the near-universal standard. All simulation frameworks for biomarker-stratified designs (Simon 2010), adaptive enrichment trials (Freidlin & Korn 2014), basket and umbrella trials (Renfro & Sargent 2017), and platform trials use direct mean moderation with explicit interaction terms. Recent simulation work on biomarker-by- treatment interaction estimation in randomized trials with prognostic covariates (Haller & Ulm 2018) likewise adopts the standard regression specification, focusing on power and bias of the interaction estimator under different covariate adjustment strategies. N-of-1 simulation studies (Zucker et al. 1997; Araujo et al. 2016; Duan et al. 2013) use hierarchical random-effects models in which individual treatment effects are drawn from a population distribution, a variant of Architecture A in which the biomarker predicts the random treatment effect through the mean structure. Methodological work on power and design for N-of-1 trials with serial correlation and carryover (Wang & Schork 2019; Schork 2022) adopts the same Architecture A specification, even though it shares senior authorship with Hendrickson et al. (2020). This suggests that the differential-correlation parameterization in the latter is a paper-specific design choice rather than a methodological convention of any laboratory or research program.

Architecture B (differential correlation via MVN) appears primarily in latent variable and structural equation modeling contexts (Rizopoulos 2012), Bayesian joint modeling frameworks, and copula-based simulation studies. In the clinical trial simulation literature specifically, the Hendrickson et al. (2020) framework appears to be unique in using differential correlation as the mechanism for biomarker-treatment interaction in an N-of-1 trial context.

This predominance of Architecture A has an important implication: the power estimates reported in the vast majority of biomarker-moderated trial simulation studies may be optimistic for biomarkers that operate through an Architecture B mechanism, because these studies do not account for the additional power loss that carryover produces when the interaction signal resides in the covariance structure rather than the mean structure.

8.3 7.3 Crossover and N-of-1 trial methodology

The crossover trial literature (Senn 2002; Jones & Kenward 2014) addresses carryover extensively but does not distinguish between architectures for biomarker-treatment interaction modeling. More recent methodological work on testing for carryover effects via mixed-effects models (Sturdevant & Lumley 2021) likewise treats carryover as a nuisance parameter to be estimated and adjusted for, without engaging with how carryover interacts with biomarker-by-treatment effect signals. The standard recommendation, namely to always include carryover in the analysis model or to use adequate washout, applies to both architectures but has different power implications under each. The finding that Architecture B is more sensitive to carryover suggests that crossover designs with short washout periods may be less suitable for detecting correlation-based biomarker interactions than previously recognized.

8.4 7.4 Precision medicine trial design

Enrichment and biomarker-stratified designs (Simon 2010; Freidlin & Korn 2014) use Architecture A exclusively for power calculations. Three families of design are particularly worth considering in light of the architectural distinction.

Adaptive enrichment trials adjust the eligible population mid-trial based on emerging evidence of a biomarker-treatment interaction. The interim power calculations that drive these adaptive decisions are universally framed in the Architecture A regression idiom. If the true biomarker mechanism is closer to Architecture B and any participants pass through washout intervals between dose-finding stages, the realized power for the interaction test will be lower than the adaptation rule expects, biasing decisions toward premature futility stopping or toward overly narrow enrichment criteria. The discrepancy is most acute when the biomarker mechanism is uncertain at the design stage, which is precisely the situation in which adaptive enrichment is most attractive.

Treatment-switching designs (most crossover, basket, and platform trials, as well as N-of-1 hybrid designs) cycle individual participants through different drug states. Under Architecture B, each washout phase erodes the BM-BR correlation, so the biomarker-treatment interaction signal in the second and subsequent treatment periods is structurally weaker than in the first. The power loss documented in Section 3 applies here directly. By contrast, parallel-group enrichment designs, in which each participant remains on a single treatment for the duration of the trial, escape this loss because there is no off-drug phase during which the correlation can decay; for these designs, Architectures A and B yield similar power estimates. The architectural choice therefore matters most in exactly the designs where the biomarker mechanism is most likely to be of clinical interest, namely those that deliberately expose each participant to both treatment states in order to estimate within-subject responsiveness.

Basket and umbrella trials (Renfro & Sargent 2017) introduce an additional layer because their power calculations average across multiple biomarker-defined arms. If even a subset of the arms operate through an Architecture B mechanism while the trial-wide power calculation assumes Architecture A throughout, the aggregated power estimates will be biased optimistically in a way that is difficult to detect from trial-level results alone. Subgroup-specific power diagnostics under both architectures should accompany the primary planning analysis whenever the underlying biology is heterogeneous across baskets.

In all three settings the practical recommendation parallels that of Section 5.3: when the biological mechanism is uncertain, run the power calculation under both architectures and treat the Architecture B estimate as the more conservative anchor for sample size planning. This recommendation is consistent with the PATH Statement (Kent et al. 2020), which itself counsels caution about data-driven discovery of treatment-by-covariate interactions and prefers risk modeling approaches when the application criteria are met.

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9 8. Conclusions

1. The choice between direct mean moderation (Architecture A) and differential correlation (Architecture B) for generating biomarker-treatment interactions in clinical trial simulations has substantive consequences for power estimates under carryover.

2. Under Architecture A, carryover reduces power only modestly, by 1 to 3% in relative terms across the three designs examined, because the proportional biomarker-drug relationship is preserved during washout. Under Architecture B the loss is strongly design-dependent: it remains modest in the crossover and hybrid designs (3 to 5%) but reaches roughly 31% in the open-label OL+BDC design, because the differential-correlation signal erodes as the drug effect wanes.

3. The choice should be guided by the hypothesized biological mechanism: Architecture A for biomarkers that directly determine drug pharmacokinetics or pharmacodynamics; Architecture B for biomarkers that statistically predict treatment response through latent subtype membership.

4. For the prazosin-PTSD application with blood pressure as the predictive biomarker, Architecture B (MVN differential correlation) is the more appropriate model, and the resulting power sensitivity to carryover should inform trial design decisions regarding off-drug period duration.

5. The existing clinical trial simulation literature uses Architecture A almost exclusively. The Hendrickson et al. (2020) MVN approach is, to the best

of our knowledge, unique in the N-of-1 trial context. This means that published power estimates for biomarker-moderated crossover and N-of-1 designs may be systematically optimistic for biomarkers that operate through a correlation-based (Architecture B) mechanism, because they do not account for the carryover sensitivity documented here.

6. Simulation studies for biomarker-moderated treatment effects should explicitly report their DGP architecture and consider sensitivity analyses under the alternative when the biological mechanism is uncertain.

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10 Conflict of interest

The authors declare no conflicts of interest.

11 Funding

This work received no specific funding.

12 Data availability statement

The data and code that support the findings of this study are openly available in the pmsimstats-ng project repository. Per-replicate Monte Carlo outputs and ADEMP-compliant cell-level summaries are versioned under `analysis/data/`; simulation drivers are at `analysis/scripts/`. Exact paths for this manuscript are listed in the Reproducibility section below.

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13 Reproducibility

This manuscript was rendered with R 4.4.x inside the project's zzcollab Docker container (image hash recorded in `Dockerfile`; package set captured in `renv.lock`). Principal packages used by the simulation pipeline are `nlme` (continuous-time linear-mixed analysis with `corCAR1` residual correlation), `parallel` (8-core `mclapply` for parameter-grid sweeps), `data.table` (the `original-extended` simulation collection), `furrr` and `purrr` (the `tidyverse` simulation collection), and `rmarkdown` plus `bookdown` for the manuscript build.

Random-number generator. The simulation driver sets `set.seed(20260507)` at the top of the replicate loop and uses `parallel::mc.set.seed` inside `mclapply`. Per-replicate seeds derive from the master seed by `+ rep_idx`.

Data and code availability. Per-replicate simulation outputs are checkpointed at `analysis/data/quick-sim/01-dgp-replicates.rds`; Morris ADEMP cell-level summaries are at the companion `01-dgp-summary.txt`. The simulation driver is at `analysis/scripts/quick-sim/01-dgp-prototype.R`. The manuscript source, bibliography, and SIM CSL file are at `analysis/report/01-dgp-mean-moderation-vs-mvn/`

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